Reflection of the experiment

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Table 1: TCAV results

Experiment	Layer	Concept Set	Control ID	TCAV Score
Concept Stability	layer2	striped_A	N/A	0.517
Concept Stability	layer3	striped_A	N/A	0.720
Concept Stability	layer2	striped_B	N/A	0.196
Concept Stability	layer3	striped_B	N/A	0.343
Control Sensitivity	layer2	striped_A	0	0.846
Control Sensitivity	layer3	striped_A	0	0.413
Control Sensitivity	layer2	striped_A	1	0.916
Control Sensitivity	layer3	striped_A	1	0.909
Control Sensitivity	layer2	striped_A	2	0.832
Control Sensitivity	layer3	striped_A	2	0.706
Layer Sensitivity	layer1	striped_A	0	0.818
Layer Sensitivity	layer2	striped_A	0	0.951
Layer Sensitivity	layer3	striped_A	0	1.000
Layer Sensitivity	layer4	striped_A	0	1.000
Layer Sensitivity	layer1	striped_A	1	0.727
Layer Sensitivity	layer2	striped_A	1	0.776
Layer Sensitivity	layer3	striped_A	1	0.315
Layer Sensitivity	layer4	striped_A	1	1.000
Layer Sensitivity	layer1	striped_A	2	0.867
Layer Sensitivity	layer2	striped_A	2	0.965
Layer Sensitivity	layer3	striped_A	2	0.972
Layer Sensitivity	layer4	striped_A	2	1.000

How stable was the TCAV signal?

From the CSV results, the TCAV signals for the striped_A and striped_B concepts show distinct patterns in stability and magnitude. In the Concept Stability experiment, both concepts exhibit increasing TCAV scores from layer2 to layer3, which might means that deeper representations capture these concepts more consistently. However, striped_B shows consistently lower TCAV scores than striped_A at both layers, suggesting that striped_A is more strongly represented in the model.

In Control Sensitivity experiment, when varying random control sets, striped_A behaves differently across layers. Across three different control sets, layer2 had a mean TCAV score of about 0.865 and a standard deviation of about 0.045 (range 0.832–0.916). In contrast, layer3 had a lower mean of about 0.676 and a much larger standard deviation of about 0.249 (range 0.413–0.909). This indicates that the signal is relatively stable at layer2, but can be highly sensitive to the sampled control set at layer3.

Which factor affected the results the most?

Among concept / control / layer, the layer produced the largest and most systematic change in the TCAV score. In the Layer Sensitivity experiment, TCAV scores generally increased with depth. For example, some layers showed strong interaction with the control set: at layer3, the score ranged from 0.315 to 1.000 depending on the control set.

The concept factor can be partially observed from the Concept Stability rows, where striped_B is consistently weaker than striped_A at both layer2 and layer3. Overall, the main conclusion is that layer dominates, while control choice can still matter for specific layers such as layer3.

What does this imply about using TCAV as an auditing signal?

These results support using TCAV as an auditing signal, but only with basic robustness checks. Because scores can shift with layer selection and occasionally react strongly to a particular control set (as seen at layer3), a single TCAV number from a single layer and single control set is not reliable enough to justify a strong causal story.

In practice, TCAV is most informative when reported as a pattern across layers and as a distribution across multiple random controls. Consistently high scores at some deeper layers suggest that the concept can be represented clearly in later representations, but other deep layers may still fluctuate (e.g., layer3 ranges from 0.315 to 1.000 across controls), indicating sensitivity and potential instability.